

1.5 Atomic structure and the periodic table

Students will be assessed on their ability to:

- a recall the definitions of relative atomic mass, relative isotopic mass and relative molecular mass and understand that they are measured relative to $1/12^{\text{th}}$ the mass of a ^{12}C atom
- b demonstrate an understanding of the basic principles of a mass spectrometer and interpret data from a mass spectrometer to:
 - i deduce the isotopic composition of a sample of an element, eg polonium
 - ii deduce the relative atomic mass of an element
 - iii measure the relative molecular mass of a compound
- c describe some uses of mass spectrometers, eg in radioactive dating, in space research, in sport to detect use of anabolic steroids, in the pharmaceutical industry to provide an identifier for compounds synthesised for possible identification as drugs
- d recall and understand the definition of ionization energies of gaseous atoms and that they are endothermic processes
- e recall that ideas about electronic structure developed from:
 - i an understanding that successive ionization energies provide evidence for the existence of quantum shells and the group to which the element belongs
 - ii an understanding that the first ionization energy of successive elements provides evidence for electron sub-shells
- f describe the shapes of electron density plots (or maps) for s and p orbitals
- g predict the electronic structure and configuration of atoms of the elements from hydrogen to krypton inclusive using $1s \dots$ notation and electron-in-boxes notation (recall electrons populate orbits singly before pairing up)
- h demonstrate an understanding that electronic structure determines the chemical properties of an element
- i recall that the periodic table is divided into blocks, such as s, p and d